

PETadex: A comprehensive database of plastic-degrading enzyme sequence diversity

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Abstract

Introduction

Plastics are a cornerstone of modern life, improving the accessibility and affordability of necessities such as medical supplies and food packaging. They are indispensable in much of the Global South, where alternative materials are scarce and prohibitively expensive [cite]. The same traits responsible for these benefits, low production cost and durability, have also made plastics a persistent, large-scale pollutant: over 400 million tonnes are produced annually, and 13 billion tonnes have accumulated since the mid-20th century [plastic production citation here]. About half of this plastic waste has been recycled or incinerated (6.3 billion tonnes), while the remaining 6.7 billion tonnes are unaccounted for in the environment [plastic pollution citation here] (Fig. 1A). This unaccounted-for waste exceeds the mass of all animals on Earth (an estimated 2.5 billion tonnes [cite animal biomass]) by 2.6-fold (Fig. 1A).

Unaccounted-for plastic waste degrades into micro- and nanoplastics (MNPs), which permeate Earth’s ecosystems, enter food chains and bioaccumulate in humans [Thompson 2024 citation here]. MNPs should be considered toxic pollutants: polystyrene MNPs fed to mice cause focal disruption of the gut epithelial barrier and inflammatory cytokine release [cite].

Despite the relative novelty of plastics, some microorganisms have adapted to break them down using plastic-degrading enzymes (PDEs). In 2016, Yoshida *et al.* isolated *Ideonella sakaiensis*, a bacterium that uses polyethylene terephthalate (PET) as its major carbon and energy source, and identified the enzyme responsible, IsPETase [cite Yoshida 2016]. PDEs such as IsPETase represent a promising solution to plastic pollution: they break plastics down into their constituent monomers, which can then be repolymerized into new plastic products or repurposed into biomass (Fig. 1B).

Existing databases of PDEs are limited in scope and diversity. The plastics-active enzymes database (PAZy) contains 495 functionally verified plastic-active enzymes [cite PAZy], most of them homologs of leaf-branch compost cutinase (LCC) [cite] or IsPETase. Sequence-based mining is a well-established method for discovering new enzymes: Danso *et al.* (2018) [cite] identified 504 candidates by searching UniProtKB and NCBI-nr, and a further 349 by mining 133 metagenomes. More recently, Erickson *et al.* (2022) [cite] screened 87 hot-spring metagenomes (~38 million sequences), characterizing 74 putative enzymes, testing 51 for activity and identifying 23 novel plastic-active enzymes. These surveys have searched only a small fraction of the sequencing data now publicly available.

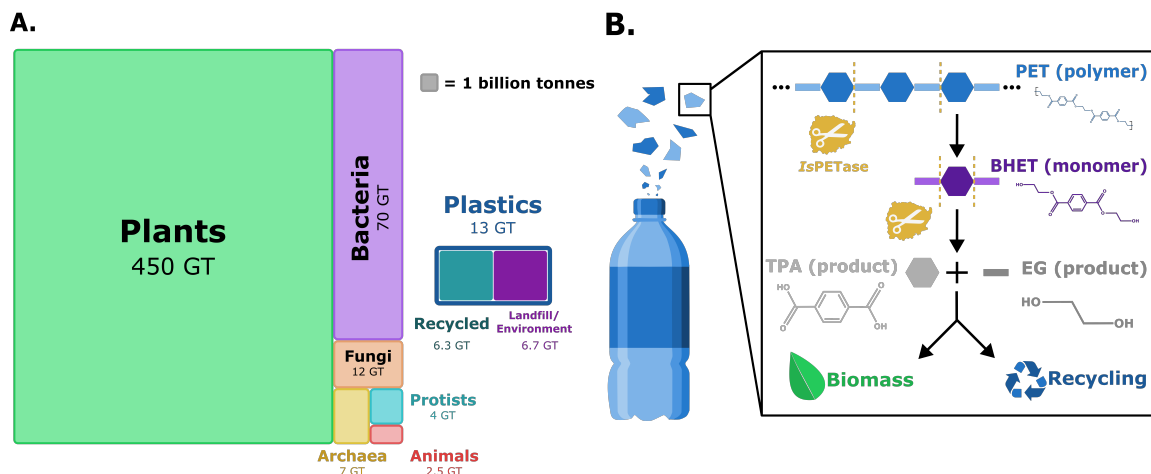


Fig. 1 Plastic waste and its enzymatic degradation. (A) Mass of plastics compared with Earth’s major groups of life (1 square = 1 billion tonnes). Of 13 Gt of plastics, 6.3 Gt has been recycled or incinerated and 6.7 Gt remains in landfill or the environment, 2.6-fold the mass of all animals (2.5 Gt). [update Fig. 1A graphic: label should read “recycled or incinerated”, not “recycled”] (B) Plastic-degrading enzymes as a route out of the plastics crisis: IsPETase hydrolyses PET to the monomer BHET, then to TPA and EG, which feed biomass or recycling.

In previous work, we built Logan [cite], an assembly of the entire Sequence Read Archive (SRA), 27.3 million metagenomic datasets, and searched it with 213 characterized PDEs, retrieving 215.7 million putative PDEs [cite PETadex v1]. We demonstrated that novel PDEs can be discovered in this dataset through operationally random selection. However, the lack of annotations prevents strategic selection of candidates for validation.

Here we present PETadex, a methodological and quantitative rebuild of our previous dataset that supports the targeted selection of PDEs. Using 411 representative PDEs as input, we searched NCBI-nr and the SRA (via the Logan assembly), filtering hits with catalytic-residue-constrained profile HMMs. The resulting dataset of [x] million enzymes carries catalytic domain boundaries and is annotated with fold assignments, predicted protein structures and the temperature and pH of the biosamples of origin. Candidates can therefore be filtered by catalytic integrity, structural fold and environmental meta-data before testing, enabling high-throughput targeted selection that can be further automated with machine learning. The enzymes, annotations and a toolkit of methods are freely available at <https://petadex.net>.

Results

Sequence and structure-based annotation of 609 known plastic-degrading enzymes

Following established workflows for the homology-based retrieval and expansion of a given set of enzymes of interest [cite Erickson, Danso, Logan etc.], we built a workflow for the retrieval of PDEs from NCBI-nr and the SRA (using the Logan assembly).

Improving upon our previous PETadex dataset, by updating the input search query and fixing several methodological limitations, we present the updated PETadex v2 dataset.

To generate this updated PETadex-260701 dataset, we created a starting search query from combining two versions of the Pazy database, the first from the original PETadex generation (pazy-240826; $n = 212$; August 26th, 2024), and a recently updated version (pazy-260701; $n = 473$; July 1st, 2026). Sourcing 136 enzymes from pazy-240826, 397 enzymes from pazy-260701 and 76 merged between both, generating a 609 enzyme union set (plasticome). Plasticome was then clustered at 90% amino acid identity to 411 cluster centroids (Fig. 2A). Predicting structures of these centroids using ESMFold2 (ESMC-6B trunk),

Fig 2A. Plasticome Composition

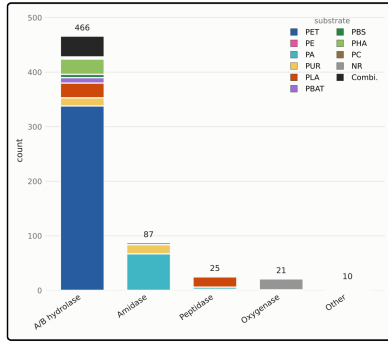


Fig 2B. PETadex Generation

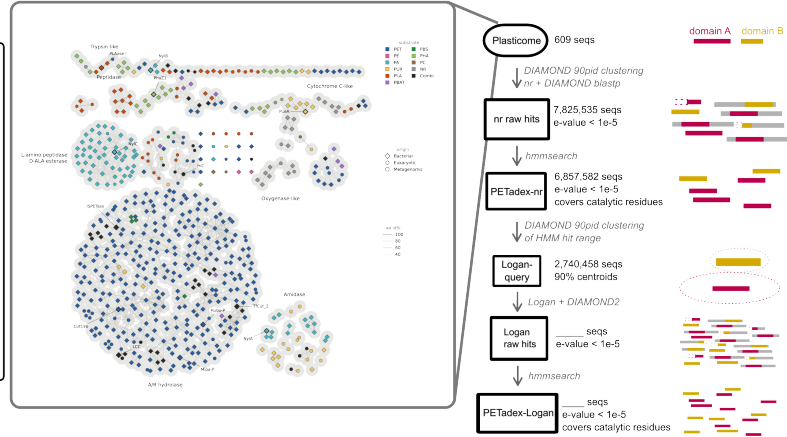


Fig 2C. PETadex Clustering

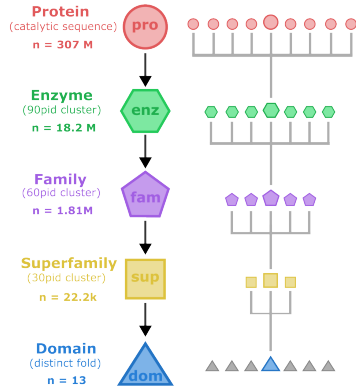


Fig 2D. Cluster Size Distribution

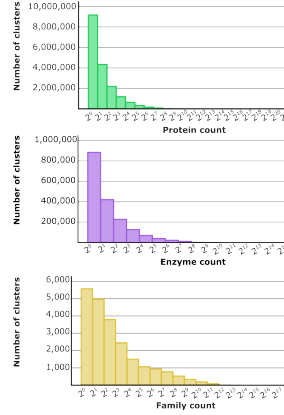


Fig. 2 The PETadex-260701 dataset. (A) Composition of the plasticome: the 609-enzyme union of the two PAZy snapshots, its collapse to 411 cluster centroids at 90% amino acid identity, the 46 graph components those centroids resolve into, and the CATH folds assigned to them, coloured by provenance. (B) Overview of PETadex generation, from the 411 input sequences through the PETadex-nr and PETadex-logan expansions. (C) Hierarchical clustering of catalytic sequences into enzymes (90% identity), families (60%), superfamilies (30%) and distinct folds. (D) Cluster size distributions at the protein, enzyme and family levels.

then searches against the CATH S40 database with Foldseek alongside searching against InterProScan resolved the proteins into 15 distinct protein domains. These centroids are further aligned into 46 graph components ($\geq 30\%$ amino acid identity and $E < 1 \times 10^{-5}$) (Fig. 2A).

Homology-based retrieval yields a dataset of 215.7 million putative plastic-degrading enzymes

To expand the dataset of known PDEs beyond the plasticome set, we first searched these 411 centroids into the NCBI nr database using DIAMOND(DIAMOND v2.2.4.184, `--very-sensitive, -k0`), recovering 42,053,081 pairwise alignments, collapsing to 8,850,583 unique sequences [cite]. Of these, 7,825,535 pass the e-value filter ($E \leq 1 \times 10^{-5}$), and further

removing plasticome self-hits returns a final count of 7,825,107 sequences (Fig. 2B).

In order to search Logan with only the catalytic domains of these plasticome homologs, we built profile HMMs to characterize the plastic-degrading domain boundaries and catalytic residues. These HMMs were built from seed MSAs (MUSCLE 5.3) of plasticome sequences together with their closest nr homologs ($E \leq 1 \times 10^{-50}$), where each member of the alignments must contain the residues involved in the catalytic mechanism, annotated from literature [cite]. The 46 graph components were further segmented using these catalytic mechanisms, resolving into 53 MSAs. The HMMs were built from the seed MSAs using a pipeline analogous to JACKhmmer [cite]. HMMs were created using HMMbuild, and then searched (HMMsearch) against close NCBI-nr homologs ($E \leq 1 \times 10^{-10}$), annotating the catalytic residues [cite]. Alignments were created from sequences that were below the e-value cut-off of 1.0E-10 and contained all of the canonical catalytic residues. These MSAs serve as the seed MSAs for the next iteration of the pipeline. While creating the next iteration’s HMMs, match and insert state annotations were propagated from the previous HMM and any column of the alignment that had an occupancy below 0.7 was declared an insert state. The pipeline was then repeated for a total of 4 iterations.

Searching these component HMMs against the 7,825,107 nr sequences yielded 7,562,397 HMM hits. Of these hits, the HMM range of 6,857,582 contained all of the positions of the catalytic residues (“complete hits”), of which 6,343,582 contained the canonical catalytic residue identities. The ranges annotated as the catalytic domains were then clustered (DIAMOND –linclust), resulting in 2,740,458 90% centroids (Fig. 2B). These centroids serve as the input query into Logan.

Expanding on the nr search, we queried these [x] million enzyme homologs across Logan (version [x]) contigs, recovering [x] billion matching sequences (Fig. 2B). Of these Logan sequences, [x] matched an nr enzyme at 90% identity, and clustering the remaining [x] at 90% aaid, yielding [x] million non-redundant enzyme homologs (Fig. 2C,D). We’ve dubbed this dataset PETadex-v2, with sub sets dubbed PETadex-plasticome (or

simply plasticome), PETadex-nr and PETadex-logan, depending on the origin. [Describe a domain from the search, can be a/B hydrolase, Amydases, etc.]

Structural and taxonomic annotation of the PETadex enzyme space

To explore the PETadex dataset, we annotated its sequences structurally and taxonomically, and linked each sequence to the metadata of the sample it was recovered from.

We first predicted structures for the 22,235 superfamily centroids (30% amino acid identity) using ESMFold2 (ESMC-6B trunk). These structures resolve both the catalytic domain and the accessory domains flanking it, which may additionally contribute to the catalytic mechanism.

We next assigned each sequence a source organism, from the NCBI taxonomy of its GenBank record (PETadex-nr) or the taxonomic composition of its source run (PETadex-logan). Of [x] PETadex-nr sequences, 1,192,350 ([x]%) carry an assignment, overwhelmingly bacterial (1,071,209; 89.8%), followed by Eukaryota (103,380; 8.7%), Archaea (17,493; 1.5%) and Viruses (268; <0.1%) [taxonomy panel removed from Fig. 3; cite a Supplementary figure or drop the reference].

Sample metadata provides planetary-scale annotation of the PETadex dataset

Each PETadex-logan sequence is taken from an SRA run, which links to an associated BioSample record, and potentially to an overarching BioProject. These records carry metadata features, such as the location, biome, host, etc., describing the sample, its collection conditions and subsequent sequencing methods. For our dataset of 307,155,746 ORFs, 256,236,322 are linked to BioSamples (83.4% coverage), drawn from 5,170,787 BioSamples (Fig. 3A).

Tracking the coverage of the metadata fields for the BioSamples, we can see that the BioSample records are most complete when describing collection date (47.5%), stated coordinates (32.0%) and isolation source (28.4%) (Fig. 3A). From the geographical coordinates of the samples, we can visualize the truly global span of PETadex

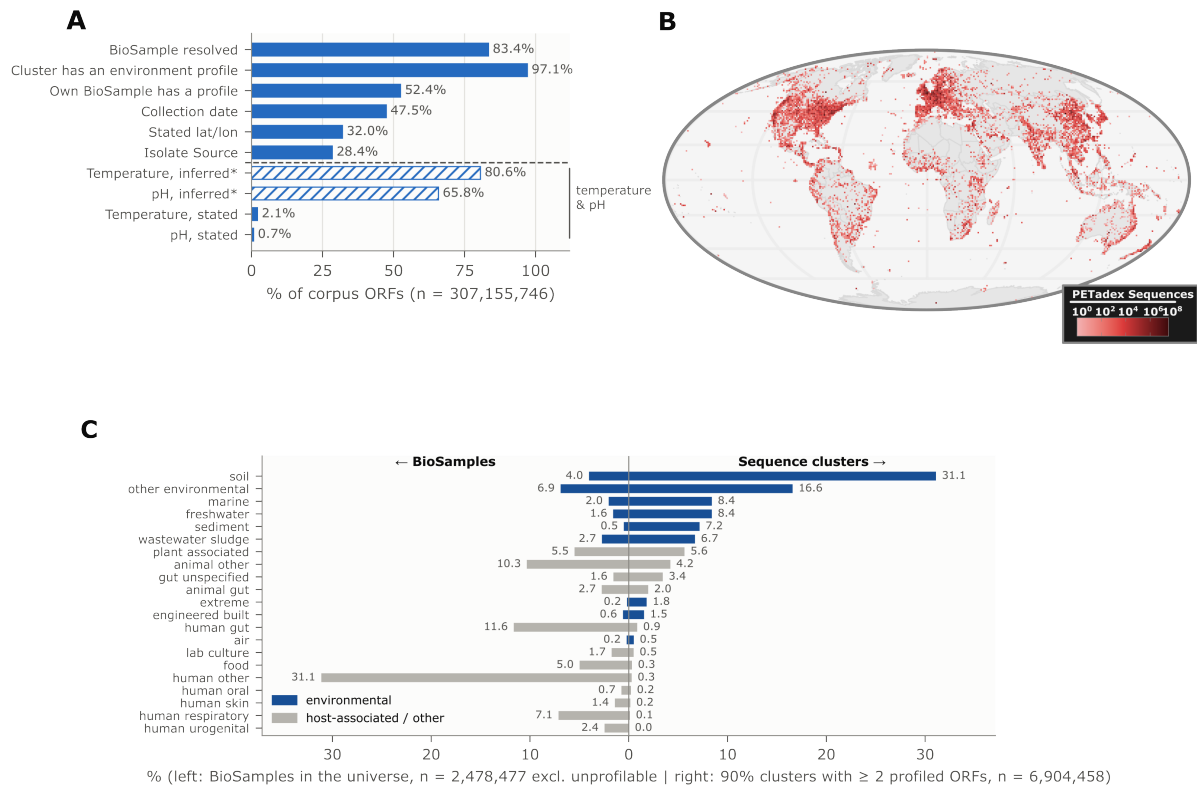


Fig. 3 Environmental metadata of the PETadex corpus. (A) Share of corpus ORFs ($n = 307,155,746$) with each metadata field. Solid bars, stated in BioSample records; hatched bars (*), inferred from BacDive optima of the taxa in each BioSample. (B) PETadex sequences by stated BioSample coordinates (log scale). (C) Environment categories of profiled BioSamples (left; $n = 2,478,477$) and of 90% clusters with ≥ 2 profiled ORFs (right; $n = 6,904,458$). Blue, environmental; grey, host-associated or other.

putative PDEs, with notable hotspots in North America, Europe and East Asia. It is important to note, however, that the span of these enzymes largely tracks sequencing runs globally, or simply put, these enzymes are found anywhere and everywhere sampling occurs. Other metadata features describing the physical conditions at collection, such as temperature and pH, are rarely documented, with stated temperature available for 2.1% of ORFs and stated pH for 0.7%. This reflects the sparse and often unreliable nature of physical condition measurements throughout the SRA.

BioSample records are also highly heterogeneous, for instance the isolation source field alone contains 78,439 distinct values. To make these

records comparable, and to allow enzyme candidates to be selected for experimental validation by their environment of origin, we classified BioSamples into 21 environment categories based on their stated environment terms, isolation source and host (Methods). Of the 2,744,599 BioSamples carrying at least one usable metadata field, 2,478,477 were assigned a category. Through their BioSample, 52.4% of the 307 million ORFs receive a category directly. To extend this coverage, we assigned each 90% sequence cluster the dominant category of its member ORFs and propagated it to members without a category of their own, raising coverage to 97.1% of ORFs (Fig. 3A). Counted by BioSample and by cluster, the categories give opposite pictures (Fig. 3C). Human-associated

categories account for 54.3% of profiled BioSamples, predominantly *human other* (31.1%) and *human gut* (11.6%), but only 1.7% of clusters with at least two profiled ORFs. Environmental categories account for 18.7% of BioSamples but 82.2% of clusters, led by soil (31.1% of clusters) and followed by other environmental sources (16.6%), marine and freshwater (8.4% each), sediment (7.2%) and wastewater sludge (6.7%). Thus, while sequencing effort in the SRA is dominated by host-associated samples, most of the sequence diversity of PETadex enzymes lies in environmental samples.

To improve the sparse metadata annotation for environmental physical conditions of temperature (2.1%) and pH (0.7%), we attempt to leverage community species composition for BioSamples. For each run, we first averaged BacDive growth optima of the taxa detected and then weighted by read abundance to obtain an approximation of sample conditions (Methods) [cite BacDive]. This proved effective at raising coverage from 2.1% to 80.6% of ORFs for temperature, and from 0.7% to 65.8% for pH (Fig. 3A). These inferred optima describes the organism within a sample, so we validated them against stated measurements for BioSamples where data was available. Abundance-weighted optima correlate with measured pH across 6,473 BioSamples (Pearson $r = 0.34$, against 0.14 without weighting) and with measured temperature across 45,741 BioSamples ($r = 0.16$, against 0.05). These samples are compressed towards neutral pH and mesophilic temperatures, and therefore do not improve on a constant in absolute error. We therefore propose that inferred optima can be applied as a means of ranking and filtering samples, but is not reliable as an estimate of *in situ* conditions, and should thus be treated as distinct from stated values.

Together these annotations support homology-based characterization of afore uncharacterized enzymes, and allow candidates to be filtered on catalytic integrity, fold and habitat prior to testing.

PETadex makes every annotation layer queryable

To make the dataset and its annotations usable for enzyme discovery, we built PETadex as an openly accessible web resource (<https://petadex.net>)

in which every annotation layer described above can be queried and cross-referenced. The database is organized around six linked entity types: sequences (307,155,746 ORFs); their cluster centroids at 90%, 60% and 30% amino acid identity; the 609 plasticome reference enzymes; the BioSamples the sequences were recovered from; the plastic substrates (polymers) that the reference enzymes act on; and kinetic measurements linking reference enzymes to substrates. Because these entities are linked, a user can traverse from any putative enzyme to its cluster centroid, from the centroid to its nearest characterized plasticome enzyme and that enzyme’s known substrates and kinetics, and from the sequence to the environmental metadata of the sample it came from.

The main entry point is sequence search. Query proteins are searched against the full PETadex corpus with DIAMOND [cite DIAMOND], which we split into [32] shards searched in parallel, giving an end-to-end latency of $\sim$ [44] s per query (Methods) [add latency vs corpus size and/or query length, ideally as a Supplementary figure]. Each search is stored with a persistent URL, so a result set can be shared or cited and later retrieved without being recomputed, which makes candidate selection from PETadex reproducible. Each hit links to a sequence page that reports its cluster memberships, closest plasticome enzyme, source organism and BioSample metadata.

Alongside search, PETadex provides four browsable layers. The ESM Atlas is an interactive embedding map of the 64,730 [60%?] family centroids, where users can locate plasticome enzymes and explore uncharacterized neighbourhoods of sequence space in ESM-2 embedding space [cite ESM-2]. Predicted ESMFold2 structures of the 22,235 superfamily centroids can be viewed in the browser from each centroid page. The Enzyme Activity layer collects 464 kinetic records for plasticome enzymes against a library of 20 polymer substrates, curated from the literature (Methods). The BioSample layer exposes the environmental metadata of each sample, including the 21 environment categories and the stated and inferred temperature and pH described above (inferred values cover 80.6% of ORFs for temperature and 65.8% for pH), with inferred values labelled separately from stated ones.

All PETadex data are openly available. Sequences, cluster assignments, annotations and metadata can be downloaded in bulk from public Amazon S3 buckets [bucket names/URLs], and the PlasticESM model and its training set are released on Hugging Face [URL]. Data are released under [licence, e.g. CC BY 4.0], and the source code for the website and search infrastructure is available on GitHub (Code availability).

A single superfamily shows that characterized enzymes sample a narrow corner of the sequence space

To show how these annotation layers combine in practice, we examined one 30% identity superfamily in detail: superfamily 6, the α/β -hydrolase superfamily (CATH 3.40.50.1820) that contains both IsPETase and LCC, along with the two further reference enzymes Cut190 and TfCut2 [cite Cut190, TfCut2]. It contains 41,871 ORFs, 32,230 of them from Logan and 9,565 from NCBI-nr, resolving into 407 families at 60% identity and 5,532 clusters at 90% identity. Of these 90% clusters, 62% are recovered only from Logan, and 129 of the 407 families contain Logan sequences alone. Characterized plasticome enzymes fall in just 40 of the 407 families, so most of the superfamily has no characterized representative at all.

We inferred a maximum-likelihood phylogeny of 563 representatives of the superfamily: the 388 family centroids, 161 characterized plasticome enzymes, the four reference enzymes and two centroids from neighbouring superfamilies (Fig. 4A; Methods). The neighbouring superfamily centroids nest inside superfamily 6 rather than outside it, and no candidate outgroup we tested fell outside the superfamily, so the tree is midpoint-rooted and the 30% identity boundary is an operational unit rather than a clade. The deep backbone is weakly supported (15% of nodes below 70% UFBoot) and we do not interpret deep branching. The superfamily is nonetheless environmentally structured rather than cosmopolitan: testing each family for enrichment of an environment category against the rest of the superfamily, counting distinct BioProjects so that a single large study cannot drive an enrichment, 82 of 130 testable families are significantly enriched for one category (one-sided Fisher, Benjamini–Hochberg $q < 0.05$), most often soil (26 families), aquatic

(17) or engineered environments (15) [this enrichment analysis is panel A of composite v2, not of the v1 figure committed here; either swap the figure or cite a Supplementary panel].

Domain architecture across the superfamily is highly uniform: the α/β -hydrolase core is present in 41,815 of the 41,871 members (99.9%) and occurs alone in 91.8% of them, and the canonical Ser-Asp-His catalytic triad is conserved in 548 of the 555 representatives present in the corpus (Fig. 4B). The remaining 7.3% of members, drawn from 47 families, carry an accessory domain [cite Pfam]. The most frequent is the type-IX secretion system C-terminal sorting domain (T9SS-CTD, PF18962; 1,662 members in 24 families, 84% of them from Logan), followed by a Ricin-B lectin (729 members) and the carbohydrate-binding module CBM2 (324). The T9SS-CTD marks a Bacteroidota radiation on the opposite side of the tree from all four reference enzymes, and is more deeply phylogenetically conserved than any environmental or sample-type trait we measured on this superfamily (consenTRAIT τ_D 1.45-fold the permuted null on the family tree, $p = 0.002$) [cite consenTRAIT], though less so than host taxonomy. Only two characterized enzymes, PET30 and PET27, sit within it [cite PET30, PET27]. Its largest family (2,287 ORFs, 97.6% from Logan, drawn from 438 BioProjects and enriched in engineered environments, mostly wastewater sludge, with freshwater samples next most common) sits at a median 46.7% identity to any characterized enzyme, making the secretion-tagged lineage a concrete example of a candidate group that is invisible to a search of characterized sequences alone.

Finally, we measured how far this diversity sits from what has been characterized (Fig. 4C,D). Logan members lie a median 61.8% identity from their nearest characterized enzyme, against 78.9% for nr members, with 32% and 11% respectively below 50% identity. This comparison is ORF-weighted, and because near-identical nr sequences pile up in a few large clusters next to characterized enzymes it overstates the difference: at 90% cluster level the medians are 56.5% (Logan-only) and 62.3% (nr-only), a gap of 6 points rather than 17, still significant (Mann–Whitney $p = 1.5 \times 10^{-7}$) but modest [decide which unit the figure and text report; composite v2 shows

the cluster-level version]. Some of this difference is expected by construction, since characterized enzymes come from cultured organisms deposited in nr. The reference enzymes themselves have strikingly few close relatives: only 41 (LCC) to 233 (Cut190) of the 41,871 members are $\geq 90\%$ identical to a reference, and an exhaustive search of all nr and PAZy ORFs together with every 60% family centroid found no closer homolog anywhere else in PETadex. Most of the superfamily is 30–50% identical to all four references. The characterized enzymes that anchor this field therefore sample a narrow corner of the sequence space their own superfamily occupies.

Discussion

PETadex complements, rather than replaces, curated resources of plastic-degrading enzymes. PAZy [cite PAZy] and PlasticDB [cite PlasticDB] catalogue experimentally characterized enzymes, and PMBD [cite PMBD] collects plastic-degrading microorganisms and genes from the literature. PETadex is several orders of magnitude larger, is derived mainly from metagenomes, and pairs each sequence with the environmental context of its source sample, while linking back to characterized enzymes through the plasticome. This scale has limits. Apart from the plasticome, PETadex entries are putative enzymes that have not been validated experimentally, and the website labels them as such [confirm the rebuild shows this on every non-plasticome page]. Environmental metadata are recorded per sample, not per ORF, so they describe where a sequence was found rather than the conditions its enzyme is adapted to. To keep the resource maintainable, PETadex is released in numbered versions (the present release is PETadex-260701), and it runs on serverless infrastructure whose costs scale with use rather than with uptime [give the maintenance plan and funding horizon, plus approximate monthly cost]. Future releases will add new annotation layers, including novel domain and TED structural-domain annotations [cite TED], feed wet-lab assay results for PETadex candidates back into the Enzyme Activity layer, and accept community submissions of characterized enzymes.

Methods

Web platform and sequence search

The PETadex website is a Next.js application hosted on Vercel [backend, database engine and hosting (v1 used AWS Lambda + RDS PostgreSQL; confirm for the rebuild)]. For sequence search, the PETadex corpus is divided into [32] DIAMOND databases (DIAMOND v[x]; [sensitivity mode, e-value cutoff, max target sequences]) that are searched in parallel on [compute service], and the per-shard hits are merged and ranked by [bit score / e-value]. Search results are written to [storage] under a unique identifier that forms the persistent result URL. Latency was benchmarked as the end-to-end time from submission to rendered results for [n] queries of [length range]. The ESM Atlas was built from ESM-2 [cite ESM-2; model size, e.g. 650M] embeddings of the family centroids [pooling, dimensionality reduction and parameters]. Kinetic records for the Enzyme Activity layer were curated from the published literature [search strategy, date range, inclusion criteria, extracted fields]. [Structure viewer: library used to render ESMFold2 models.]

Superfamily case study

Members of superfamily 6 were exported from the PETadex clustering tables with their 90% and 60% cluster assignments and origin, and characterized enzymes were mapped onto them with DIAMOND (ultra-sensitive) at $\geq 95\%$ identity and $\geq 80\%$ coverage. The tip set of 563 sequences comprises the 388 family centroids at 60% identity, the four reference enzymes as their own exact ORFs, one characterized member per 90% cluster containing a plasticome enzyme (161), 8 plasticome enzymes that are superfamily-like ($\geq 50\%$ identity to a member) but absent from the corpus, and 2 centroids of neighbouring superfamilies. Sequences were aligned with FAMSA [cite FAMSA] and trimmed with ClipKIT (kpic-smart-gap) [cite ClipKIT], retaining 988 of 2,603 columns, and the tree was inferred with IQ-TREE using ModelFinder (WAG+F+I+R10 by BIC) and 1,000 ultrafast bootstrap replicates [cite IQ-TREE, ModelFinder, UFBoot]. Because every candidate outgroup nested inside the superfamily,

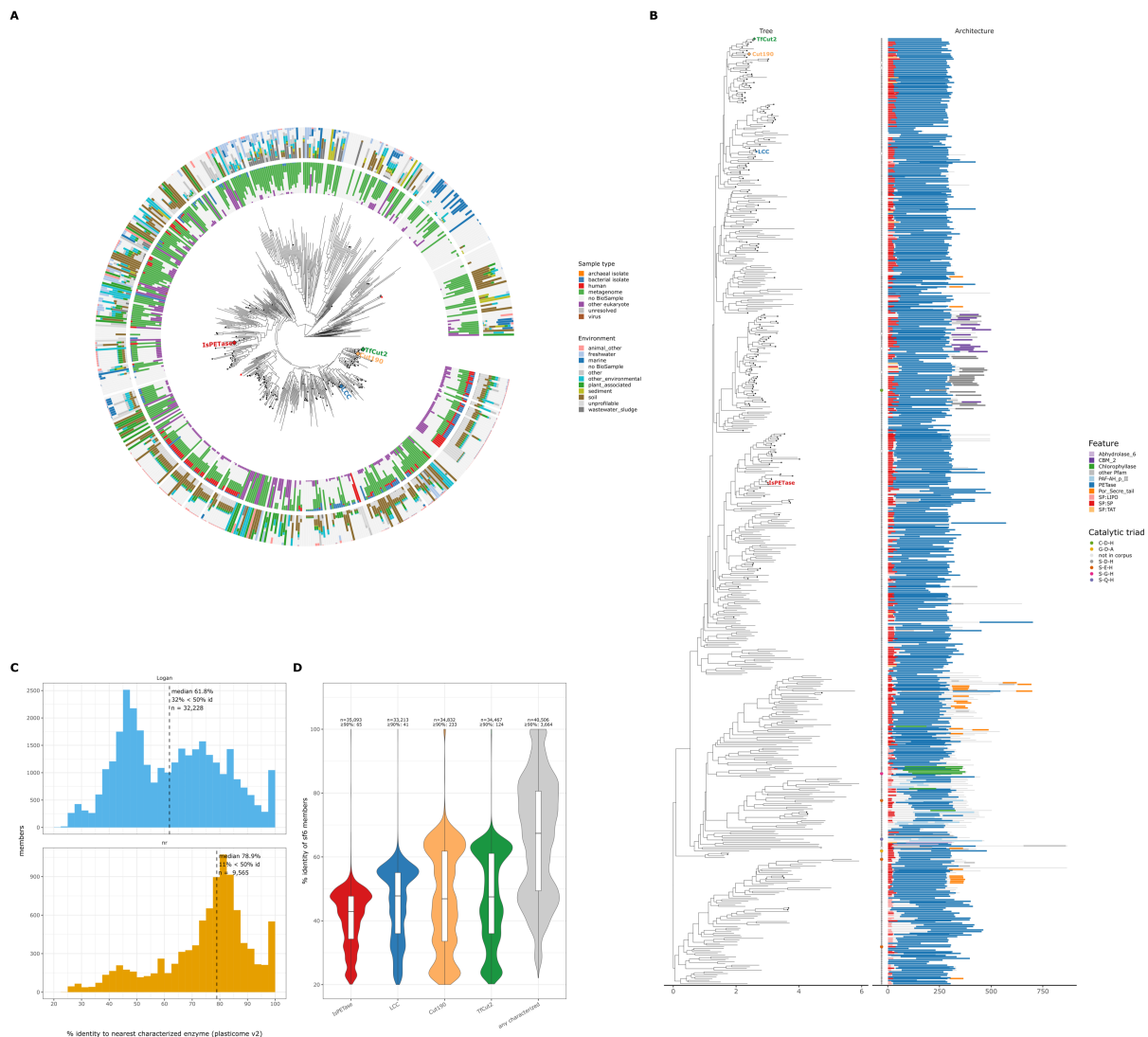


Fig. 4 Characterized PET hydrolases sample a narrow corner of the IsPETase/LCC superfamily, most of whose diversity is recovered only from Logan. PETadex superfamily 6 (30% identity cluster) contains 41,871 ORFs, 77% of them from Logan assemblies of SRA runs, in 407 families at 60% identity. The superfamily is an operational unit, not a clade: neighbouring superfamilies nest within it. **(A)** Maximum-likelihood phylogeny (IQ-TREE, WAG+F+I+R10, midpoint-rooted) of 563 representatives: 388 family centroids (60% identity), 161 characterized plasticome enzymes (filled dots), 8 characterized sequences absent from PETadex (open dots), the four reference enzymes IsPETase, LCC, Cut190 and TfCut2 (diamonds), and 2 neighbouring-superfamily centroids (red triangles). The deep backbone is weakly supported and is not interpreted. Rings, inner to outer: sample type and environment category of each family's BioSamples. **(B)** The same tree drawn linearly, with the domain architecture of each representative: Pfam domains, SignalP 6.0 signal peptides, and the catalytic triad (dots; Ser-Asp-His in 548 of the 555 representatives present in the corpus). **(C)** Identity of each member to its nearest characterized enzyme (plasticome v2). Logan members: median 61.8%, 32% below 50% ($n = 32,228$). nr members: median 78.9%, 11% below 50% ($n = 9,565$). nr is enriched near characterized enzymes by construction. **(D)** Identity of every member to each reference enzyme, and to any characterized enzyme. Only 41 (LCC) to 233 (Cut190) members are $\geq 90\%$ identical to a reference.

the tree is midpoint-rooted. Domains were annotated with Pfam 37 (`hmmsearch --cut_ga`, overlapping hits resolved greedily by score) and signal peptides with SignalP 6.0 [cite Pfam, HMMER,

SignalP]; signal-peptide calls are reported for nr members only, because 99% of Logan ORFs are cut from contigs and do not begin at a methionine. Catalytic triads were taken from the profile-HMM

annotation and confirmed against the untrimmed alignment. Identity to the nearest characterized enzyme used DIAMOND with `-k 10`, taking the best bit score [state sensitivity mode and coverage filter used for the reported numbers]. Environment enrichment per family was tested with a one-sided Fisher exact test on distinct BioProjects against the rest of the superfamily, in families with ≥ 5 profiled units, with Benjamini–Hochberg correction. Phylogenetic conservation of the T9SS-CTD was assessed with Fitch parsimony and consen-TRAIT against 1,000 label permutations [cite consenTRAIT, and state which tree each statistic was computed on].

Supplementary information.

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Declarations

- Funding
- Competing interests
- Ethics approval and consent to participate
- Consent for publication
- Data availability
- Materials availability
- Code availability
- Author contribution

References